

Robust & Stochastic Control

MEAM 5170

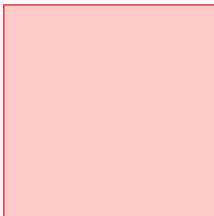
Last Time: Trajectory Optimization

Here is a general form of discrete-time trajectory optimization:

$$\begin{aligned} \min_{x_{0:T}, u_{0:T-1}} \quad & c_T(x_T) + \sum_{t=0}^{T-1} c(x_t, u_t) \\ \text{s.t.} \quad & x_{t+1} = f(x_t, u_t), \quad x_0 = x_{\text{init}}, \end{aligned}$$

A Quadrotor Example

Say we want to find a minimum length trajectory for a quadrotor to get to point A to point B:



Question: *what could go wrong?*

Defining a Stochastic Control System

Previously:

$$x_{t+1} = f(x_t, u_t), \quad x_0 = x_{\text{init}}$$

Now:

$$x_{t+1} = f(x_t, u_t, w_t), \quad x_0 \sim p(x_0),$$

where w_t is a random process.

Continuous vs Discrete

We can talk about stochastic systems in either continuous or discrete time. For simplicity, this lecture mostly focuses on discrete time.

In continuous time, you deal with **stochastic differential equations**, and have to worry about Itô's lemma, which describes how the chain rule works for SDE's.

Example: Stochastic LTI

Consider:

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad w_t \sim \mathcal{N}(0, \Sigma)$$

For some noise covariance $\Sigma = \Sigma^\top \succ 0$

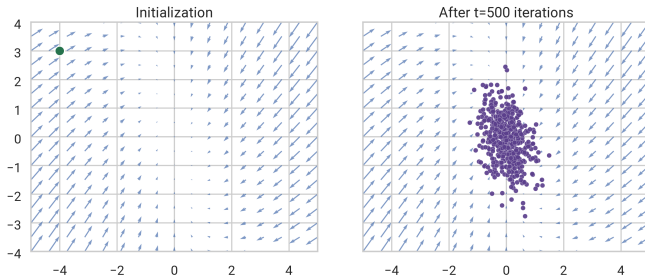
Note: if x_0 is a Gaussian and u_t is an affine function of x_t , then all x_t are Gaussian.

Let $u_t = Kx_t$ such that $\rho(A + BK) < 1$.

Question: Does the system have a fixed point at $x = 0$?

Stationary Distributions

For stochastic systems, it often makes more sense to talk about a *stationary distribution* as opposed to a stationary point.



Question: For the STLl system, if $x_0 = 0$, what happens to x_t as $t \rightarrow \infty$?

$$x_{t+1} = (A + BK)x_t + w_t, \quad w_t \sim \mathcal{N}(0, \Sigma) \text{ i.i.d.}$$

Example: Polytope Uncertainty and LTI

Consider:

$$x_{t+1} = (A + BK)x_t + w_t, \quad w_t \in \mathcal{W},$$

where $\mathcal{W} \subset \mathbb{R}^n$ is a convex polytope and $x_0 = 0$.

The *Minkowski sum* of two sets is

$$\mathcal{X} \oplus \mathcal{W} = \{x + w : x \in \mathcal{X} \ w \in \mathcal{W}\}$$

Also define $A\mathcal{X} = \{Ax : x \in \mathcal{X}\}$ for matrix A and set $\mathcal{X}$.

Then our set-valued dynamics are:

$$\mathcal{X}_{t+1} = (A + BK)\mathcal{X}_t \oplus \mathcal{W}$$

We can track the convex polytope $\mathcal{X}_t$ by its vertices $\{v_x^{(1)}, \dots, v_x^{(m)}\}$

Zonotopes

Things are easier if our sets are *Zonotopes*, defined:

$$\mathcal{Z}(c, G) = \left\{ c + \sum_{i=1}^m z_i g_i : z_i \in [-1, 1] \right\}$$

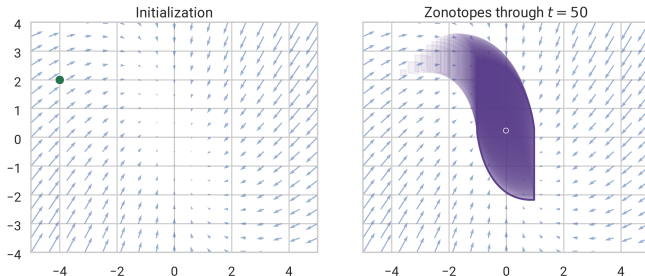
Then, the Minkowski sum becomes:

$$\mathcal{Z}(c, G) \oplus \mathcal{Z}(c', G') = \mathcal{Z}(c + c', [G, G'])$$

So, if $\mathcal{X}_t = \mathcal{Z}(c_x, G_x)$ and $\mathcal{W} = \mathcal{Z}(c_w, G_w)$, we get:

$$\mathcal{X}_{t+1} = \mathcal{Z}\left((A + BK)c_x + c_w, [(A + BK)G_x, G_w]\right)$$

Example: Zonotope evolution



Takeaway: *We can think about propagating sets through dynamics to model uncertainty*

Example: Manipulator Equations with Uncertainty

Recall:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = Bu$$

We might have uncertainty about physical parameters (e.g. mass, friction). We can represent those by ξ :

$$M(q, \xi)\ddot{q} + C(q, \dot{q}, \xi)\dot{q} + g(q, \xi) = Bu$$

Then we would get the system:

$$\begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix} = \begin{bmatrix} \dot{q} \\ M(q, \xi)^{-1} (Bu - C(q, \dot{q}, \xi)\dot{q} - g(q, \xi)) \end{bmatrix}$$

where our random process, $w_t = \xi$ is a random constant (over time).

Disturbances vs Uncertainty about Parameters

Stochastic system:

$$x_{t+1} = f(x_t, u_t, w_t), \quad x_0 \sim p(x_0),$$

We might want to separate random parameters ξ that remain constant over time:

$$x_{t+1} = f(x_t, u_t, w_t, \xi), \quad x_0 \sim p(x_0)$$

Where ξ is a random variable

Note: *Random parameters can be wrapped into the state by creating an augmented state (but then your controller loses full observability)*

Robust vs Stochastic Control

$$x_{t+1} = f(x_t, u_t, w_t), \quad x_0 \sim p(x_0),$$

Let $J(\pi; w_{0:T-1})$ be a cost function over a policy based on $w_{0:T-1}$.

Let $\pi \in \Pi$ be policy, which in the case of trajectory optimization may be parameterized by the individual control inputs $u_0, u_1, \dots, u_{T-1}$.

In general, **stochastic control** aims to optimize the expected objective:

$$\min_{\pi \in \Pi} \mathbb{E}_w J(\pi; w_{0:T-1})$$

Whereas **robust control** aims to optimize the worst-case objective:

$$\min_{\pi \in \Pi} \max_{w_{0:T-1} \in W} J(\pi; w_{0:T-1})$$

Ex: Stoch. Trajectory Optimization via Sampling

Consider a stochastic trajectory optimization problem:

$$\begin{aligned} \min_{X_{0:T}, u_{0:T-1}} \quad & \mathbb{E} \left[c_T(X_T) + \sum_{t=0}^{T-1} c(X_t, u_t) \right] \\ \text{s.t.} \quad & X_{t+1} = f(X_t, u_t, W_t), \quad X_0 = x_{\text{init}}, \end{aligned}$$

an easy way to approximate this is to sample $\{w_t^{(i)}\}$ and solve:

$$\begin{aligned} \min_{x_{0:T}^{(i)}, u_{0:T-1}} \quad & \frac{1}{n} \sum_{i=1}^n \left[c_T(x_T^{(i)}) + \sum_{t=0}^{T-1} c(x_t^{(i)}, u_t) \right] \\ \text{s.t.} \quad & x_{t+1}^{(i)} = f(x_t^{(i)}, u_t, w_t^{(i)}), \quad x_0^{(i)} = x_{\text{init}} \end{aligned}$$

Question: *why might sharing the u_t between particles be too conservative?*

Example: Robust TO with Finite Uncertainty Set

Let's presume our uncertainty is drawn from a finite set, $\{\xi^{(1)}, \dots, \xi^{(n)}\}$, then we can do a similar thing:

$$\begin{aligned} \min_{x_{0:T}^{(i)}, u_{0:T-1}} \quad & \max_{i \in 1, \dots, n} \left\{ c_T(x_T^{(i)}) + \sum_{t=0}^{T-1} c(x_t^{(i)}, u_t) \right\} \\ \text{s.t.} \quad & x_{t+1}^{(i)} = f(x_t^{(i)}, u_t, \xi^{(i)}), \quad x_0^{(i)} = x_{\text{init}} \end{aligned}$$

Question: *why do we have to assume the uncertainty set is finite for this? Why can't we sample?*

Example: Tube Trajectory Optimization

Let's go back to a bounded uncertainty set $\mathcal{W}$, and consider linearized dynamics around a nominal trajectory and error feedback.

We want to optimize a *tube* around it so that we are guaranteed to remain in the tube. We pick an invariant set $\mathcal{E}$ and can formulate:

$$\begin{aligned} \min_{\bar{x}_{0:T}, \bar{u}_{0:T-1}, \alpha_{0:T}} \quad & c_T(\bar{x}_T^{(i)}) + \sum_{t=0}^{T-1} c(\bar{x}_t, \bar{u}_t) + \sum_{t=1}^T \gamma \alpha_t \\ \text{s.t.} \quad & \bar{x}_{t+1} = A_t \bar{x}_t + B_t \bar{u}_t \quad \bar{x}_0 = x_{\text{init}} \\ & \alpha_{t+1} \mathcal{E} \supseteq (A_t + B_t K_t)(\alpha \mathcal{E}) \oplus \mathcal{W} \quad \alpha_t \geq 0 \end{aligned}$$

If $\mathcal{E} = \{x \in \mathbb{R} : \|x\| \leq 1\}$ and $\mathcal{W}$ has radius r_w , the tube dynamics constraint becomes:

$$\alpha_{t+1} \geq \|A + BK\|_2 \alpha + r_w$$

Example: Tube Trajectory Optimization

TODO: image

Chance Constraints

We don't always just care about the objective; we may also want *constraints* to be satisfied (think about the quadrotor).

However, a hard constraint like:

$$g(x_t, u_t) \leq 0$$

might not be feasible in stochastic systems (e.g. x_t has Gaussian noise injected)

Instead, we can express a *chance constraint* as:

$$P[g(x_t, u_t) \leq 0] \geq 1 - \alpha$$

How to Actually Solve Chance Constraints

TODO: this

- sampling
- propagate a parametric distribution (e.g. Gaussian) and put quantile constraints
- Show example of optimization problem

Stochastic Value Function

- state assumptions
- derive the bellman equation

Robust Value Function

- state rectangularity
- derive the bellman equation

Example: Stochastic LQR

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad w_t \sim \mathcal{N}(0, \Sigma)$$

Consider the cost $c(x, u) = x^\top Qx + u^\top Ru$ for $Q = Q^\top \succeq 0$, $R = R^\top \succ 0$. Then, our value function is:

$$V(x) = \min_u \mathbb{E}_w [x^\top Qx + u^\top Ru + \gamma V(Ax + Bu + w)]$$

Let's assume our optimal policy is $u_t = Kx_t$ and our value function is $V(x) = x^\top Sx + c$ for some $S = S^\top \succ 0$ and c . Then, we get:

$$S = Q + \gamma A^\top SA - \gamma^2 A^\top SB(R + \gamma B^\top SB)^{-1} B^\top SA$$

$$c = \frac{\gamma}{1 - \gamma} \mathbb{E}_w [w^\top Sw] = \frac{\gamma}{1 - \gamma} \text{tr}(S\Sigma)$$

Question: *what happens if $\gamma = 1$?*

When $\gamma = 1$

TODO: this

Observability

Our current formulation,

$$x_{t+1} = f(x_t, u_t, w_t)$$

doesn't give us uncertainty about state from partial-observability.
This requires introducing observations:

$$y_t = h(x_t, u_t, v_t)$$

where v_t is a stochastic process handling measurement noise.

Example: LQG

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad w_t \sim \mathcal{N}(0, \Sigma)$$

We can extend our Stochastic LTI system to have uncertain observations:

$$y_t = Cx_t + v_t, \quad v_t \sim \mathcal{N}(0, \Omega)$$

Then, we can arrive at a *Kalman filter*.

Adding a Kalman filter in the loop with LQR gives us LQG:
Linear-Quadratic-Gaussian

H_∞ Control

Consider a system with disturbances $w(t)$ and cost-relevant outputs $z(t)$:

$$\dot{x}(t) = f_\pi(x(t), w(t)), \quad z(t) = h_\pi(x(t), w(t))$$

where f_π, h_π depend on the chosen policy π .

Define a norm over a time-varying function:

$$\|x(\cdot)\|_{L_2} = \sqrt{\int_0^T x(t)^\top x(t) dt}$$

The H_∞ problem is then:

$$\min_{\pi \in \Pi} \max_{\|w\|_{L_2} \neq 0} \frac{\|z(\cdot)\|_{L_2}}{\|w(\cdot)\|_{L_2}}$$

Lyapunov Functions as Certificates in H_∞ Control

Suppose we want to build a certificate of our objective:

$$\max_{\|w\|_{L_2} \neq 0} \frac{\|z(\cdot)\|_{L_2}}{\|w(\cdot)\|_{L_2}} \leq \gamma.$$

This is equivalent to saying the following holds for all $\|w\|_{L_2}$:

$$\|z(\cdot)\|_{L_2}^2 - \gamma^2 \|w(\cdot)\|_{L_2}^2 \leq 0.$$

We can actually certify this by finding a Lyapunov function $V(x)$ satisfying:

$$\dot{V}(x) + z(t)^\top z(t) - \gamma^2 w(t)^\top w(t) \leq 0.$$

For all $w(\cdot)$.

Example: Domain Randomization in RL

Reinforcement learning with domain randomization tries to minimize some cost:

$$J(\theta) = \mathbb{E}_{\xi} \mathbb{E}_{\tau \sim \pi_{\theta}, \xi} \left[\sum_{t=1}^T \gamma^t c(x, u) \right]$$

where ξ are random domains (friction, etc.)

Interestingly, domain randomization can some times result in a better policy on the nominal dynamics than not doing domain randomization:

TODO: image

Other Measures of Risk: VaR and CVaR

Let J be a random variable denoting some sort of cost (e.g. the trajectory cost over a random w_t). Maybe we don't want to just focus on $\mathbb{E}J$.

The value-at-risk at level α is a quantile:

$$\text{VaR}_\alpha(J) = \min\{z : \mathbb{P}(J \leq z) \geq \alpha\}$$

The conditional value-at-risk is the expected tail cost:

$$\text{CVaR}_\alpha(J) = \mathbb{E}[J \mid J \geq \text{VaR}_\alpha(J)]$$

TODO: image

(Manipulation) Funnels

TODO: this

Distributionally Robust Control

TODO: this